AxC on G (facts + axioms)

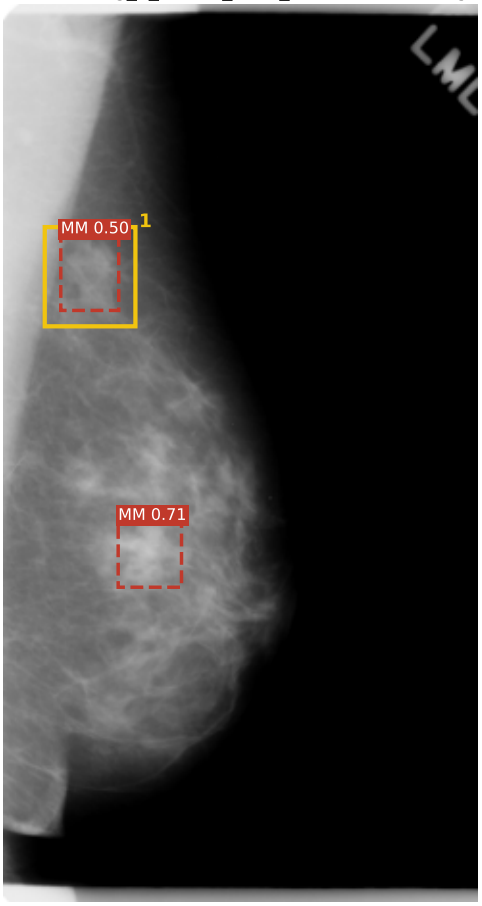
run v5_G_AxC_geo_e50 best epoch 11 validation mAP@50 0.229

```
Every page: two validation images. Yellow = annotated lesion (number = lesion id).
Solid box = the detection matched to the lesion (highest score with IoU >= 0.5);
dashed = other detections with score >= 0.3. Next to the image, per lesion:
GT = annotation (class = biopsy pathology of the image; descriptors from the CBIS CSV),
PRED = what the model says for the matched box; AXIOMS = rules of the lesion's family whose
antecedent holds, with the truth of the rule on the model's own predictions (ok >= 0.5).

Concept evaluation on ALL validation lesions (matched detections):
   lesions: 438
   detected (IoU >= 0.5): 0.589
   class accuracy: 0.740
   malignancy accuracy: 0.640
   mass_shape accuracy: 0.570
   mass_shape macro-F1: 0.242
   mass_shape majority baseline: 0.570
   mass_margin accuracy (per value): 0.785
   mass_margin macro-F1: 0.000
   mass_margin majority baseline: 0.785
   mass_density accuracy: 0.654
   mass_density macro-F1: 0.569
   mass_density majority baseline: 0.562
   calc_morph accuracy (per value): 0.770
   calc_morph macro-F1: 0.176
   calc_morph majority baseline: 0.792
   calc_distr accuracy: 0.575
   calc_distr macro-F1: 0.146
   calc_distr majority baseline: 0.575
```

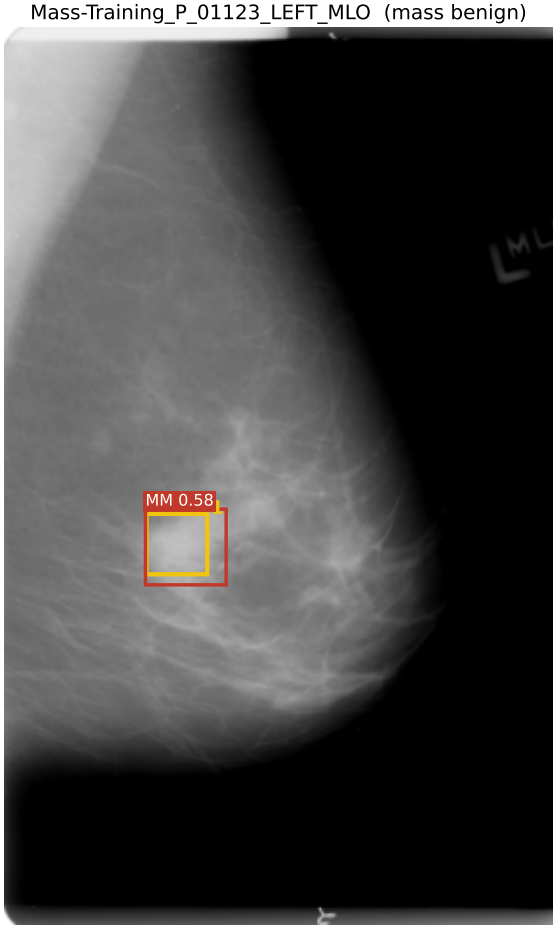
AxC on G (facts + axioms)

Mass-Training_P_01052_LEFT_MLO (mass benign)




```
Lesion 1
GT   mass benign
     shape: irregular;  margin: indistinct;  density: low
PRED not detected (no box with IoU >= 0.5)
```

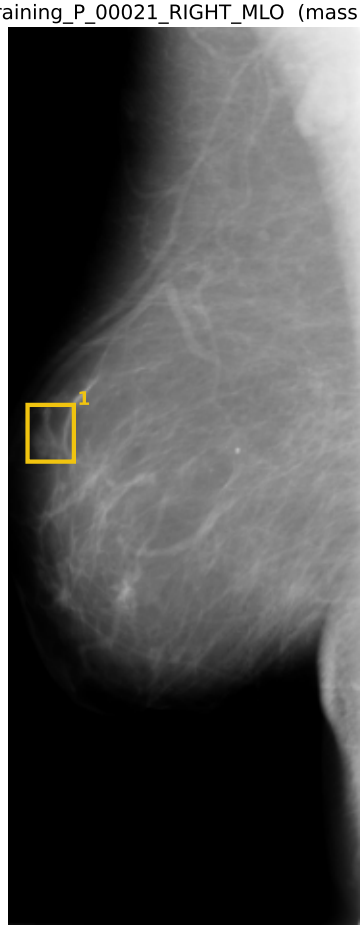
Mass-Training_P_01123_LEFT_MLO (mass benign)




```
Lesion 1
GT   mass benign
     shape: oval;  margin: circumscribed + indistinct;  density: high
PRED mass malignant 0.58 (malignant 0.61, box score 0.58)
     shape: oval 0.60;  margin: none >= 0.5 (top: circumscribed 0.31);
       density: low 0.57
AXIOMS that apply (antecedent >= 0.3):
     psi3 (mass circumscribed/obscured -> benign): 0.63 ok
     psi6 (mass irregular -> malignant): 0.88 ok
     psi9 (mass high density -> malignant): 0.83 ok
```

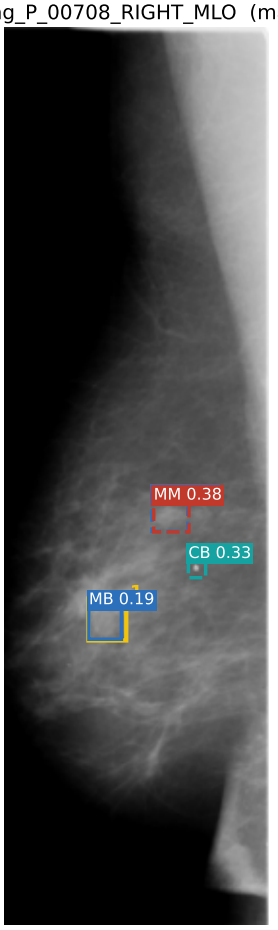
AxC on G (facts + axioms)

Mass-Training_P_00021_RIGHT_MLO (mass benign)




```
Lesion 1
GT   mass benign
     shape: oval;  margin: circumscribed;  density: low
PRED not detected (no box with IoU >= 0.5)
```

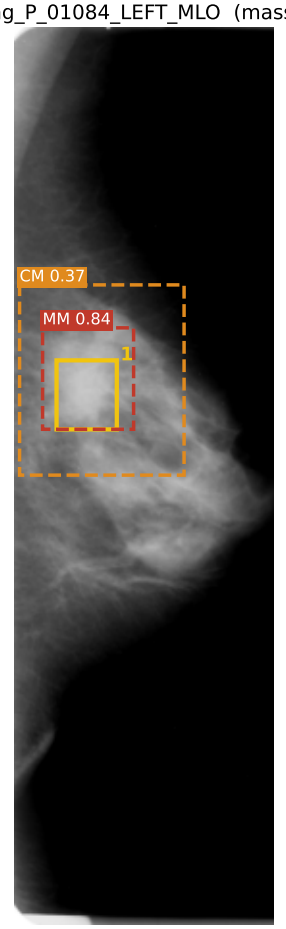
Mass-Training_P_00708_RIGHT_MLO (mass benign)




```
Lesion 1
GT   mass benign
     shape: oval;  margin: circumscribed;  density: low
PRED mass benign 0.19 (malignant 0.05, box score 0.19)
     shape: oval 0.81;  margin: none >= 0.5 (top: circumscribed 0.29);
       density: low 0.90
AXIOMS that apply (antecedent >= 0.3):
     psi3 (mass circumscribed/obscured -> benign): 0.65 ok
```

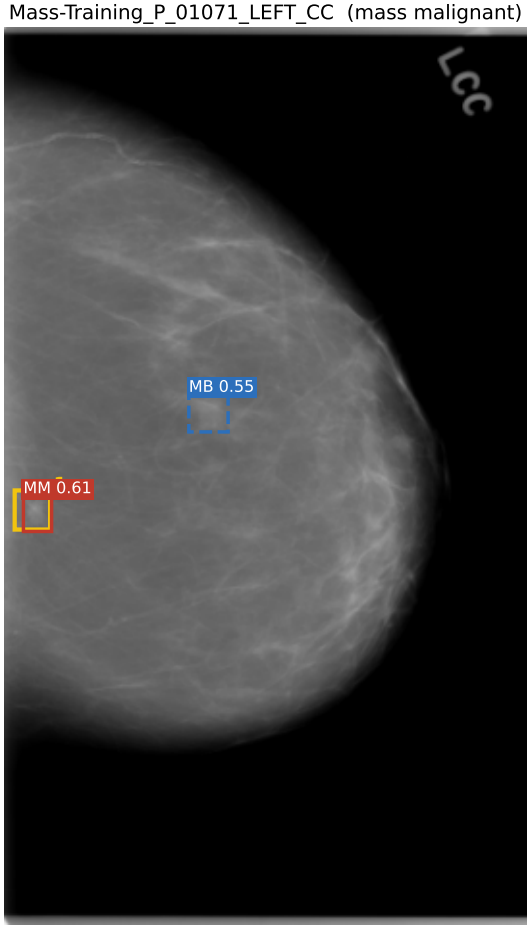
AxC on G (facts + axioms)

Mass-Training_P_01084_LEFT_MLO (mass malignant)




```
Lesion 1
GT   mass malignant
     shape: irregular;  margin: obscured + spiculated;  density: high
PRED not detected (no box with IoU >= 0.5)
```

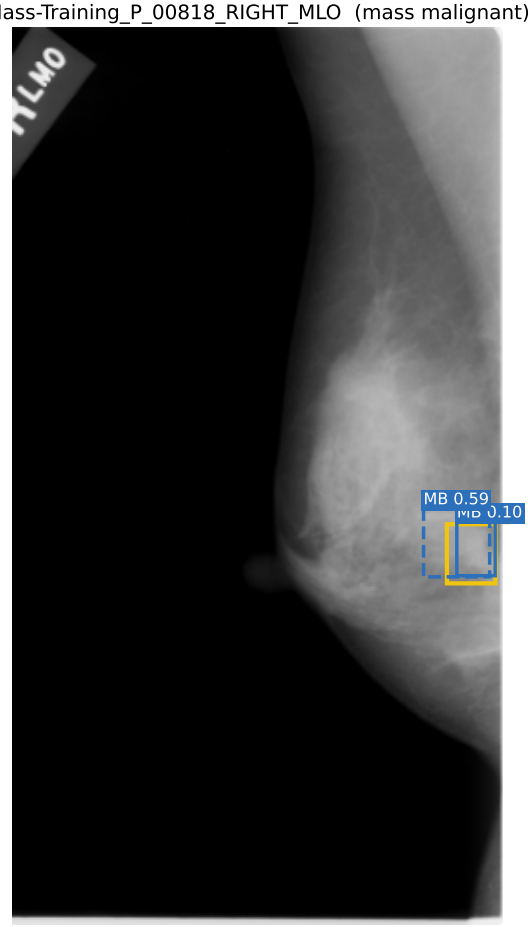
Mass-Training_P_01071_LEFT_CC (mass malignant)




```
Lesion 1
GT   mass malignant
     shape: irregular;  margin: spiculated;  density: low
PRED mass malignant 0.61 (malignant 0.63, box score 0.61)
     shape: oval 0.69;  margin: none >= 0.5 (top: circumscribed 0.31);
       density: low 0.77
AXIOMS that apply (antecedent >= 0.3):
     psi3 (mass circumscribed/obscured -> benign): 0.62 ok
```

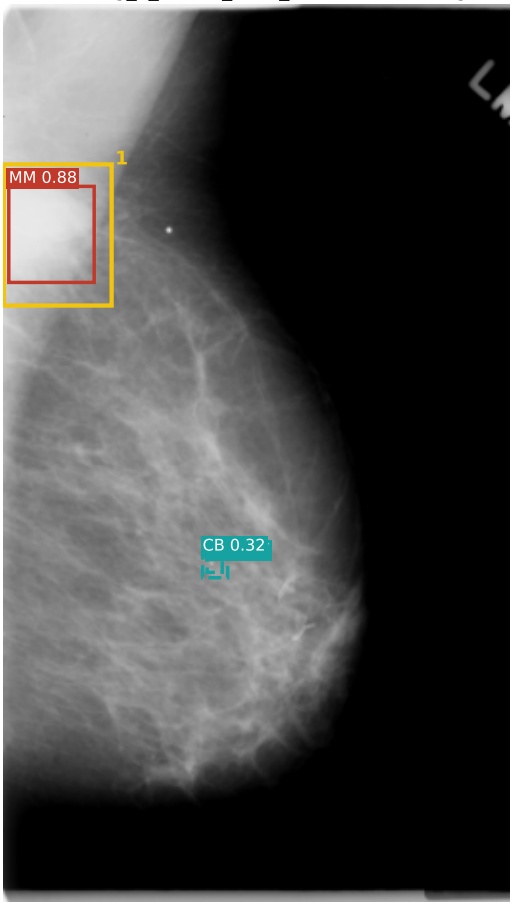
AxC on G (facts + axioms)

Mass-Training_P_00818_RIGHT_MLO (mass malignant)




```
Lesion 1
GT   mass malignant
     shape: irregular;  margin: spiculated;  density: low
PRED mass benign 0.10 (malignant 0.06, box score 0.10)
     shape: oval 0.76;  margin: none >= 0.5 (top: circumscribed 0.31);
       density: low 0.92
AXIOMS that apply (antecedent >= 0.3):
     psi3 (mass circumscribed/obscured -> benign): 0.58 ok
```

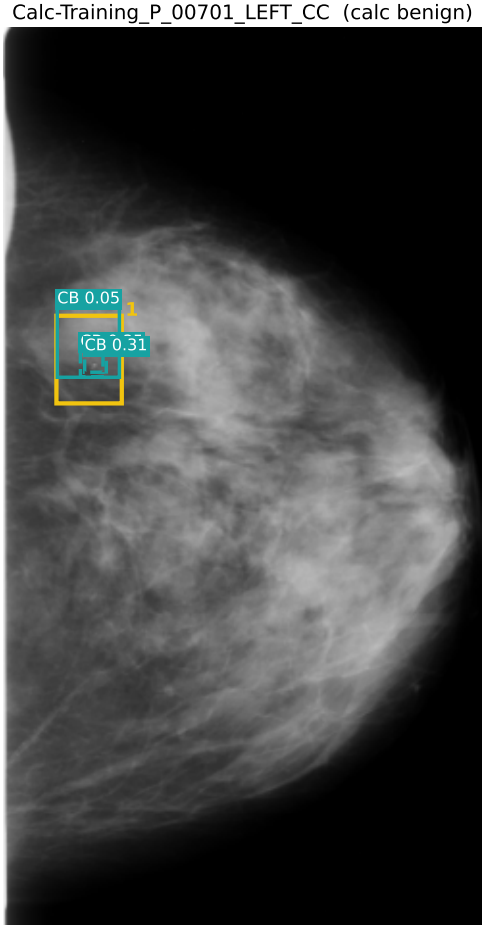
Mass-Training_P_01775_LEFT_MLO (mass malignant)




```
Lesion 1
GT   mass malignant
     shape: oval;  margin: microlobulated;  density: high
PRED mass malignant 0.88 (malignant 0.91, box score 0.88)
     shape: oval 0.59;  margin: none >= 0.5 (top: spiculated 0.26);
       density: high 0.64
AXIOMS that apply (antecedent >= 0.3):
     psi3 (mass circumscribed/obscured -> benign): 0.63 ok
     psi6 (mass irregular -> malignant): 0.97 ok
     psi9 (mass high density -> malignant): 0.94 ok
```

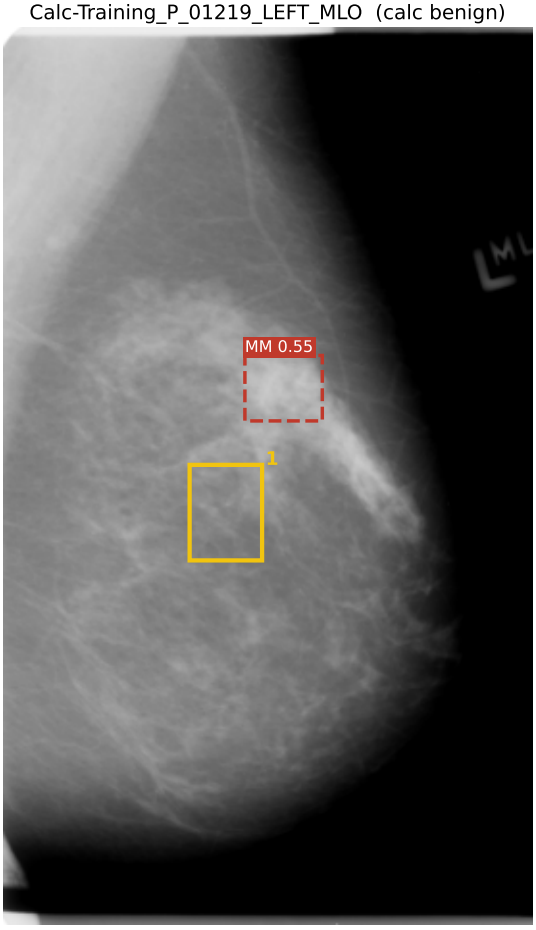
AxC on G (facts + axioms)

Calc-Training_P_00701_LEFT_CC (calc benign)




```
Lesion 1
GT   calc benign
     morphology: pleomorphic;  distribution: clustered
PRED calc benign 0.05 (malignant 0.04, box score 0.05)
     morphology: none >= 0.5 (top: pleomorphic 0.50);  distribution:
       clustered 0.80
AXIOMS that apply (antecedent >= 0.3):
     psi8 (pleomorphic, not regional/diffuse -> malignant): 0.57 ok
```

Calc-Training_P_01219_LEFT_MLO (calc benign)




```
Lesion 1
GT   calc benign
     morphology: pleomorphic;  distribution: clustered
PRED not detected (no box with IoU >= 0.5)
```

AxC on G (facts + axioms)

Calc-Training_P_00780_RIGHT_MLO (calc benign)



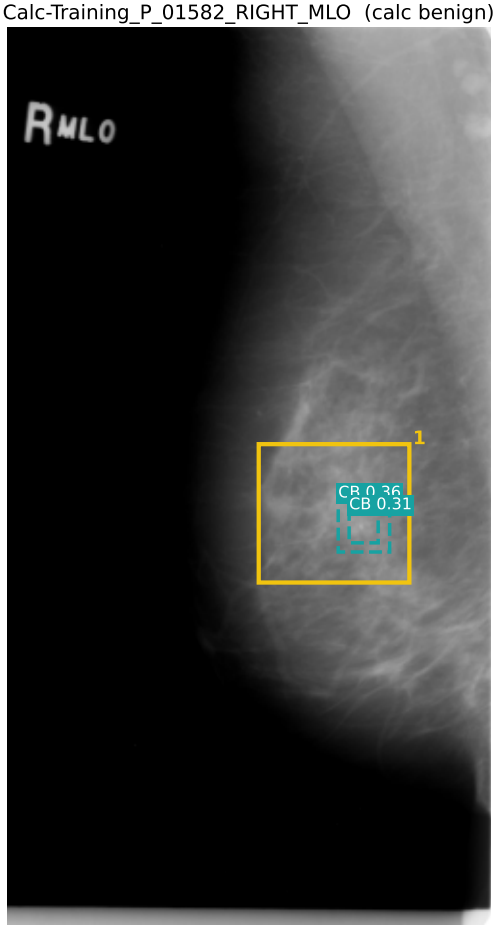

```
Lesion 1
GT   calc benign
     morphology: typically benign;  distribution: —
PRED not detected (no box with IoU >= 0.5)

Lesion 2
GT   calc benign
     morphology: typically benign;  distribution: —
PRED not detected (no box with IoU >= 0.5)

Lesion 3
GT   calc benign
     morphology: typically benign;  distribution: —
PRED not detected (no box with IoU >= 0.5)

(+ 2 more lesions not listed)
```

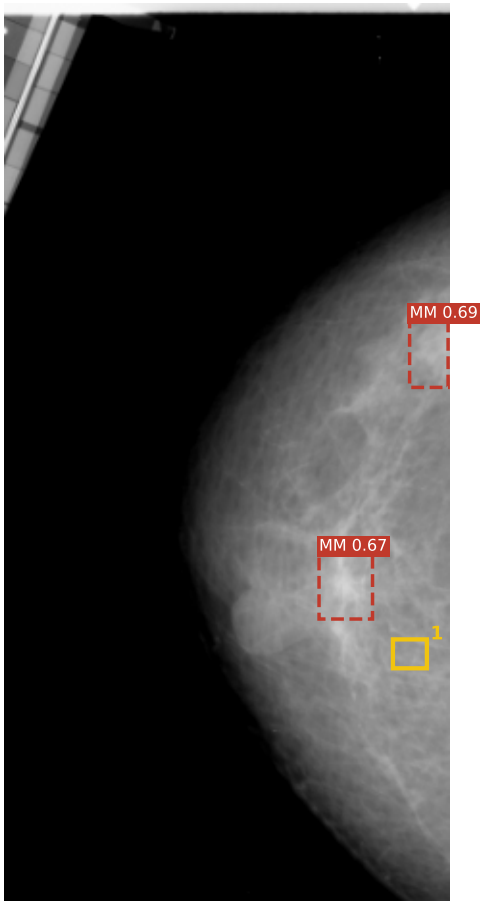
Calc-Training_P_01582_RIGHT_MLO (calc benign)




```
Lesion 1
GT   calc benign
     morphology: amorphous;  distribution: clustered
PRED not detected (no box with IoU >= 0.5)
```

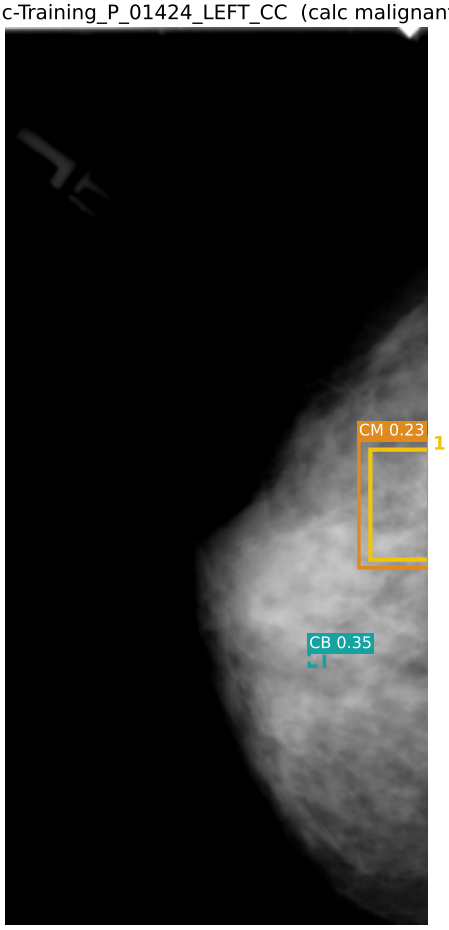
AxC on G (facts + axioms)

Calc-Training_P_00735_LEFT_CC (calc malignant)




```
Lesion 1
GT   calc malignant
     morphology: pleomorphic;  distribution: clustered
PRED not detected (no box with IoU >= 0.5)
```

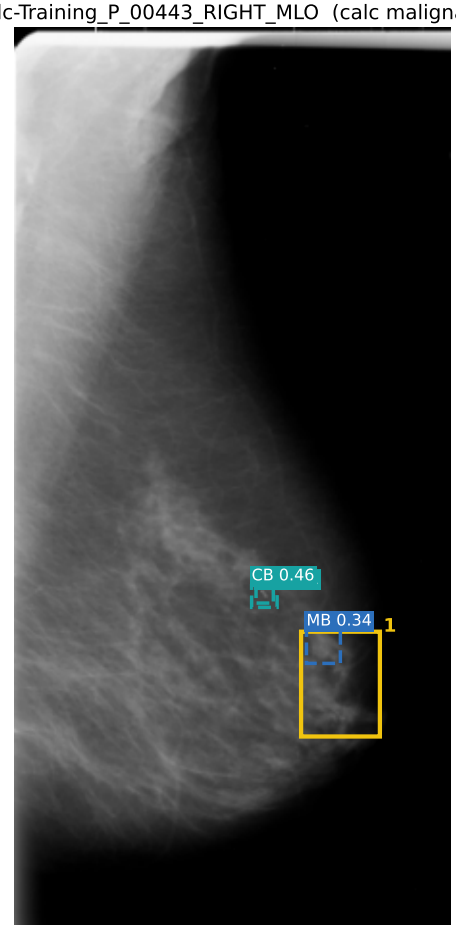
Calc-Training_P_01424_LEFT_CC (calc malignant)




```
Lesion 1
GT   calc malignant
     morphology: pleomorphic;  distribution: clustered
PRED calc malignant 0.23 (malignant 0.24, box score 0.23)
     morphology: none >= 0.5 (top: pleomorphic 0.49);  distribution:
       clustered 0.84
AXIOMS that apply (antecedent >= 0.3):
     psi8 (pleomorphic, not regional/diffuse -> malignant): 0.66 ok
```

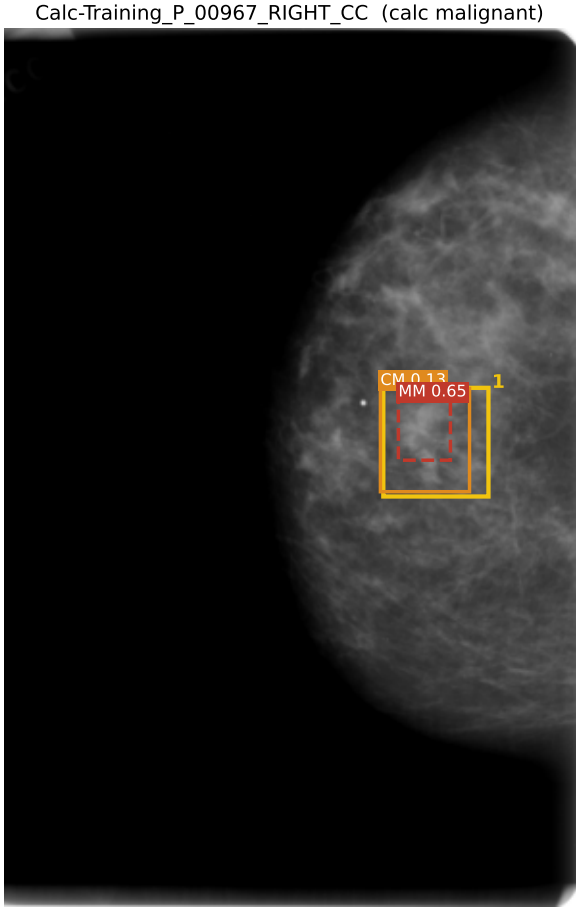
AxC on G (facts + axioms)

Calc-Training_P_00443_RIGHT_MLO (calc malignant)




```
Lesion 1
GT   calc malignant
     morphology: fine linear branching;  distribution: linear
PRED not detected (no box with IoU >= 0.5)
```

Calc-Training_P_00967_RIGHT_CC (calc malignant)




```
Lesion 1
GT   calc malignant
     morphology: pleomorphic;  distribution: clustered
PRED calc malignant 0.13 (malignant 0.19, box score 0.13)
     morphology: none >= 0.5 (top: typically benign 0.41);  distribution:
       clustered 0.82
AXIOMS that apply (antecedent >= 0.3):
     psi8 (pleomorphic, not regional/diffuse -> malignant): 0.70 ok
```